CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Case 20-2023: A 52-Year-Old Man with a Solitary Fibrous Tumor and Hypoglycemia

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PRESENTATION OF CASE

Dr. Gregory M. Cote: A 52-year-old man with a metastatic solitary fibrous tumor was admitted to this hospital because of recurrent episodes of hypoglycemia.

The patient had been in his usual state of health until approximately 1.5 years before the current admission, when dysuria and pain in the lower abdomen and left leg developed after a motor vehicle accident. At another hospital, magnetic resonance imaging (MRI) reportedly revealed an incidental 14-cm necrotic mass in the pelvis, anterior to the bladder. Biopsy of the mass was reportedly diagnostic of a solitary fibrous tumor.

The patient was referred to the multidisciplinary oncology clinic of this hospital. Computed tomography (CT) of the chest, performed for staging, revealed two 9-mm nodules in the left lung. The patient was treated with a 3-month course of dacarbazine chemotherapy and concurrent radiation therapy targeting the pelvic mass, and then he underwent surgical resection of the pelvic tumor, including partial cystectomy and placement of a ureteral stent. Pathological examination of the resected specimen (Fig. 1) confirmed the diagnosis of spindle-cell sarcoma with features consistent with a dedifferentiated solitary fibrous tumor. Immunohistochemical staining of the resected specimen revealed signal transducer and activator of transcription 6 (STAT6) and CD34 in the tumor cells. Stereotactic radiation therapy for the lung nodules was also administered.

However, metastatic disease progressed in the lungs, chest, bladder, and right leg. Over the subsequent 12-month period, the patient underwent serial treatment with four different chemotherapeutic regimens: pazopanib, doxorubicin with olaratumab, gemcitabine with docetaxel, and vinorelbine. Palliative radiation therapy for a painful leg metastasis was also provided.

Eleven days after the initiation of vinorelbine chemotherapy and 30 days before the current admission, the patient was admitted to this hospital with fever (temperature, 38.3°C), neutropenia (absolute neutrophil count, 770 per microliter; reference range, 1800 to 7700), and pneumonia in the left lower lobe.

Dr. Reece J. Goiffon: CT of the chest and abdomen (Fig. 2A), performed before and

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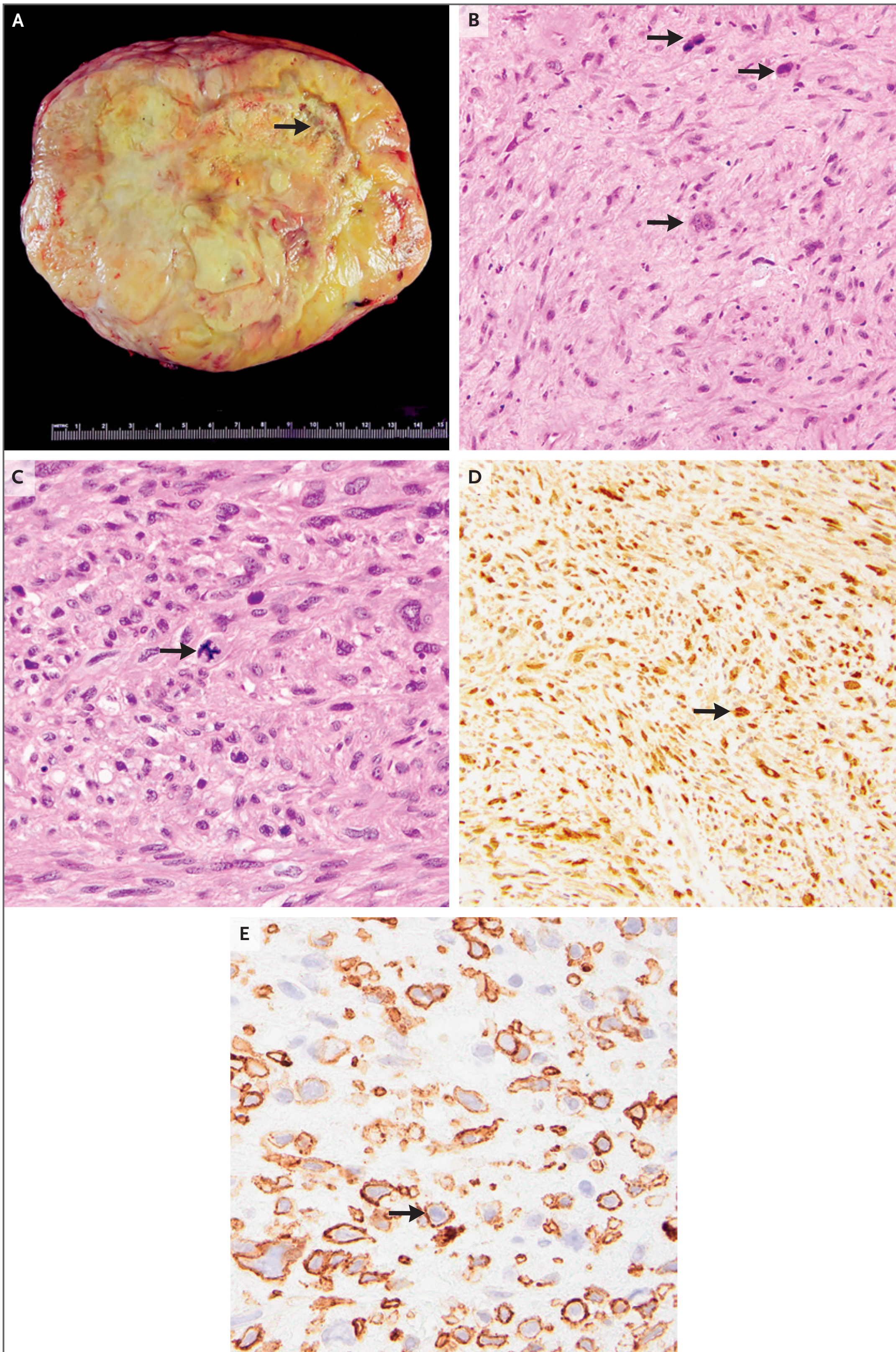
A
B
C
D
E

Figure 1 (facing page). Specimen of the Resected Mass.
The gross specimen (Panel A) shows a well-encapsulated and focally necrotic mass (arrow). Hematoxylin and eosin staining (Panels B and C) shows a high-grade spindle-cell sarcoma (Panel B, arrows) with a high level of mitotic activity (Panel C, arrow). Immunohistochemical stains for signal transducer and activator of transcription 6 (STAT6) and CD34 (Panels D and E, respectively) are diffusely positive in the tumor cells (arrows).

after the administration of intravenous contrast material, showed known metastatic involvement of the lungs, bladder, retroperitoneum, peritoneum, skeletal muscles, and subcutaneous fat. There was no metastatic involvement of the adrenal glands, pancreas, or liver.

Dr. Cote: Empirical treatment with cefepime was administered. Over the next several days, the neutropenia and fever resolved and the pneumonia abated. On the seventh hospital day, the patient was discharged home with a follow-up visit scheduled in the oncology clinic.

Eleven days after hospital discharge and 12 days before the current admission, the patient

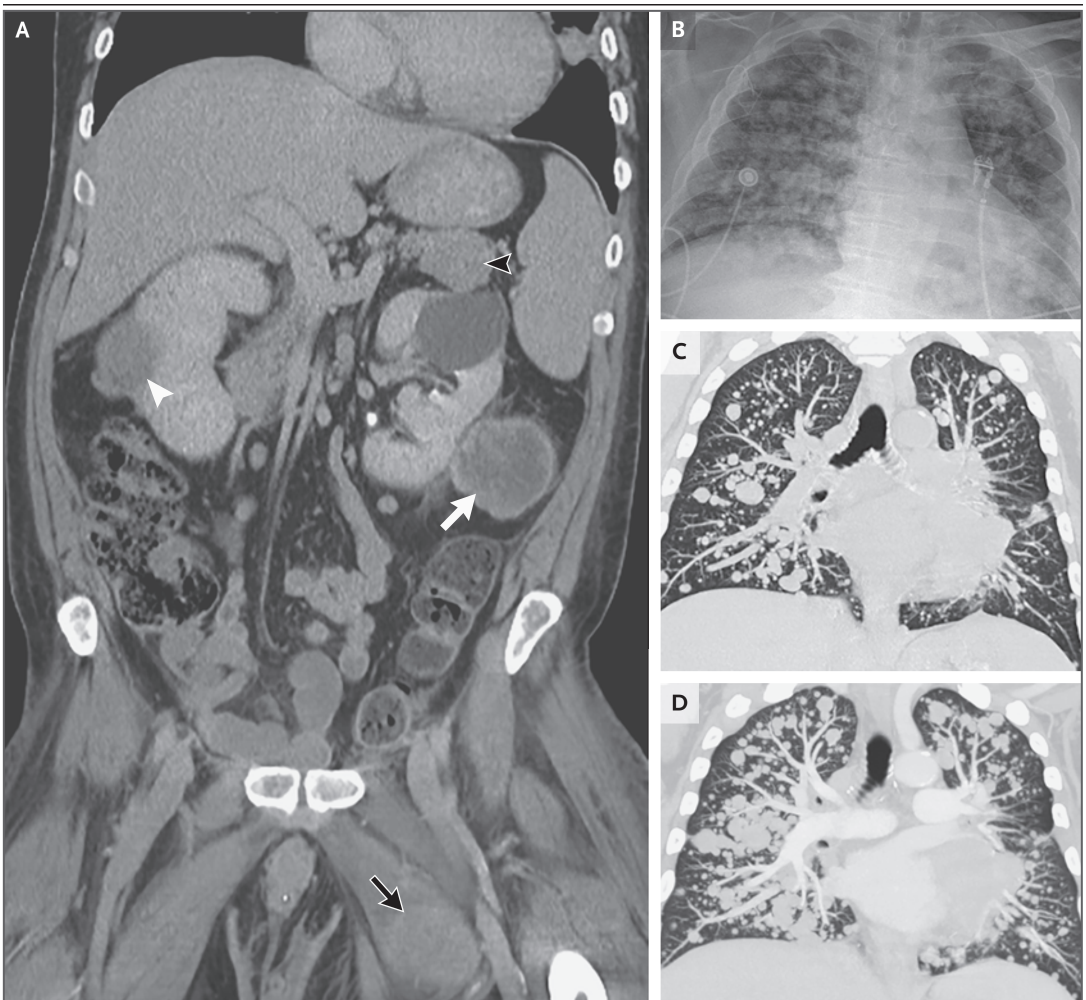

Figure 2. Imaging Studies.
A coronal, contrast-enhanced CT image of the abdomen and pelvis obtained 1 month before the current admission (Panel A) shows metastases involving the right kidney (white arrowhead), retroperitoneal fat (white arrow), peritoneal space (black arrowhead), and musculature of the left leg (black arrow). A portable chest radiograph obtained on the current admission (Panel B) shows abundant pulmonary metastases. Coronal, maximum-intensity-projection CT images of the chest obtained 1 month before the current admission (Panel C) and on the current admission (Panel D) show an increase in the size and quantity of pulmonary metastases over the 1-month period.

attended a scheduled outpatient visit for an infusion. He reported weakness and appeared confused. A fingerstick glucose level was 47 mg per deciliter (2.6 mmol per liter). Supplemental oral and intravenous glucose was administered, and his mental status improved, but fingerstick glucose levels remained below 70 mg per deciliter (3.9 mmol per liter). Review of fingerstick glucose measurements obtained during the patient's recent hospitalization showed multiple levels ranging from 50 to 60 mg per deciliter (2.8 to 3.3 mmol per liter).

The patient was admitted to the oncology unit of this hospital for the management of hypoglycemia. Laboratory test results from the second hospital admission are shown in Table 1. While the patient was in the hospital, he began to have diaphoresis that correlated with episodes of hypoglycemia, for which he received orange juice and dextrose. On evaluation by a dietitian, he was found to be meeting anticipated caloric requirements. The patient was discharged home on the third hospital day; the blood glucose level was 55 mg per deciliter (3.1 mmol per liter) before discharge.

Three days later and 6 days before the current admission, the patient attended the scheduled follow-up visit in the oncology clinic. Laboratory test results are shown in Table 1. Treatment with trabectedin was initiated, along with pegfilgrastim and a 3-day tapering course of dexamethasone. The day after the dexamethasone course was completed, the patient reported increased fatigue and cough. Two days later, he was admitted to the oncology unit because of weakness and diaphoresis; a fingerstick glucose level was 49 mg per deciliter (2.7 mmol per liter).

On the current admission, review of systems was notable for fatigue; pain associated with tumors involving the back, shoulders, and right leg; numbness of the right leg; the use of an indwelling Foley catheter for 2 months owing to urinary retention; dull chest pain and intermittent dyspnea that had persisted for 4 months; and constipation. The patient had no recent anorexia, weight change, fever, chills, orthopnea, hemoptysis, light-headedness, dizziness, palpitations, syncope, nausea, or vomiting.

The patient's medical history was notable for transfusion-dependent anemia, type 2 diabetes mellitus (for which he had received metformin, although not in the previous 6 months), hypertension, a colonic tubular adenoma, a renal cyst, osteoarthritis, chronic low-back pain, and depression. He had undergone an appendectomy in the remote past. Medications included oral gabapentin, cyanocobalamin, tamsulosin, and finasteride; transdermal fentanyl; and oral acetaminophen, clonazepam, melatonin, polyethylene glycol, and oxycodone as needed. He had no access to insulin or other hypoglycemic medications. There were no known adverse drug reactions.

The patient had formerly worked as a laborer and had been receiving disability benefits since the motor vehicle accident. He had previously consumed six beers per week and smoked one pack of cigarettes per day, but he had stopped drinking alcohol and smoking tobacco 1 year earlier. There was no history of other substance use. His family history of cancer included melanoma in his father, thyroid cancer in his sister, ovarian cancer in his maternal grandmother, leukemia in his maternal aunt, and sarcoma in his paternal uncle. His mother had hypertension and dyslipidemia, and a brother had diabetes. The patient had an adult child, who was healthy.

The tympanic temperature was 37.0°C, the heart rate 92 beats per minute, the blood pressure 148/87 mm Hg, the respiratory rate 20 breaths per minute, and the oxygen saturation 98% while the patient was breathing ambient air. Multiple tender, palpable masses were present, including masses on the left shoulder, neck, and left upper leg. A venous-access port was in place in the chest wall on the right side. The abdominal incision was well healed. A Foley catheter was in place, and there was pink-tinged urine. A faint rash was present on the lower abdomen at the site of previous radiation therapy. Edema was present in both legs. Laboratory test results from the current hospital admission are shown in Table 1.

Dr. Goiffon: A chest radiograph (Fig. 2B) showed abundant metastatic pulmonary nodules. CT of the chest (Fig. 2C and 2D), performed after the administration of intravenous contrast material, showed that the nodules had increased in size and quantity over a period of 1 month. There was no evidence of pulmonary embolism.

Dr. Cote: An electrocardiogram showed sinus rhythm, a leftward axis, nonspecific ST-segment and T-wave changes, and intraventricular conduction delay.

On the first hospital day, although the patient

Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	12 Days before Current Admission, Second Admission Day 1	11 Days before Current Admission, Second Admission Day 2	6 Days before Current Admission, Follow-up Visit	Current Admission Day 1	Current Admission Day 2
Hemoglobin (g/dl)	13.5–17.5	7.5	6.8	8.0	7.9	7.4
Hematocrit (%)	41.0–53.0	23.7	21.7	25.0	24.8	23.6
White-cell count (per μl)	4500–11,000	6350	5390	6740	11,120	6330
Platelet count (per μl)	150,000–400,000	106,000	85,000	87,000	91,000	74,000
Sodium (mmol/liter)	135–145	134	136	137	139	139
Potassium (mmol/liter)	3.4–5.0	3.3	3.4	4.3	4.0	3.8
Chloride (mmol/liter)	100–108	97	100	101	106	105
Carbon dioxide (mmol/liter)	23–32	25	23	26	23	21
Urea nitrogen (mg/dl)	8–25	3	3	6	4	3
Creatinine (mg/dl)	0.60–1.50	0.44	0.32	0.39	0.40	0.50
Glucose (mg/dl)	70–110	68	45	96	49	145
Phosphorus (mg/dl)	2.6–4.5	—	—	—	3.6	3.8
Calcium (mg/dl)	8.5–10.5	8.8	8.7	8.7	9.8	9.0
Albumin (g/dl)	3.3–5.0	2.8	—	3.0	3.1	—
Alanine aminotransferase (U/liter)	10–55	7	—	6	10	—
Aspartate aminotransferase (U/liter)	10–40	22	—	20	18	—
Alkaline phosphatase (U/liter)	45–115	65	—	72	72	—
Total bilirubin (mg/dl)	0.0–1.0	0.6	—	0.5	0.6	—
Insulin (μIU/ml)	2.6–25.0	—	<0.2	—	—	—
C-peptide (ng/ml)	1.1–4.4	—	<0.1	—	—	—
β-Hydroxybutyrate (mmol/liter)	<0.4	—	<0.1	—	—	—
Cortisol, morning (μg/dl)	5.0–25.0	—	13.5	—	—	—
Repeat cortisol, after cosyntropin (μg/dl)	—	—	21.7	—	—	—
Lactate (mmol/liter)	0.5–2.2	—	—	—	0.9	—
Troponin T (ng/ml)	<0.03	—	—	—	<0.01	—

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for phosphorus to millimoles per liter, multiply by 0.3229. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for cortisol to nanomoles per liter, multiply by 27.59. To convert the values for lactate to milligrams per deciliter, divide by 0.1110.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

was receiving a continuous infusion of dextrose, five bolus injections of dextrose and six glasses of orange juice were needed to increase the blood glucose level to more than 100 mg per deciliter (5.6 mmol per liter). On the morning of the second hospital day, the blood glucose level was 42 mg per deciliter (2.3 mmol per liter). Treatment with oral prednisone was initiated.

A diagnosis was made.

DIFFERENTIAL DIAGNOSIS

Dr. Tara E. Soumerai: This 52-year-old man with a solitary fibrous tumor that had metastasized to the lungs, chest, bladder, legs, and retroperitoneum presented with recurrent and refractory hypoglycemia. He had undergone extensive treatment with several chemotherapeutic regimens and radiation therapy, in addition to pelvic tumor resection. He had a history of type 2 diabetes mellitus, but he had not received metformin for 6 months before the current admission. Imaging studies showed evidence of widely metastatic cancer, but there was no metastatic involvement of the adrenal glands, pancreas, or liver.

SOFT-TISSUE SARCOMA

Before I construct a differential diagnosis for this patient's hypoglycemic syndrome, I will review the epidemiology of soft-tissue sarcoma in adults, particularly solitary fibrous tumors. Sarcoma is a term that encompasses any cancer that arises from connective tissue, including bone, smooth muscle, skeletal muscle, tendons, fat, and other structures. Thus, sarcoma can arise from essentially any organ in the body. The arms and legs are the most common sites in adults, followed by the thorax and abdomen.[1] Liposarcoma is the most common type of soft-tissue sarcoma in adults, followed by leiomyosarcoma, undifferentiated pleomorphic sarcoma, and gastrointestinal stromal tumors.[2] Sarcoma is very rare, with 4 cases per 100,000 persons diagnosed in the United States each year.[3] As with most cancers, the initial workup for suspected soft-tissue sarcoma includes staging imaging — typically MRI of the affected body part for surgical planning, as well as CT of the chest, with or without CT of the abdomen and pelvis, for ruling out distant metastases.[4] Biopsy is subsequently performed. The cornerstone of treatment for nonmetastatic sarcoma is surgical resection. Radiation therapy, chemotherapy, or both may be included before or after surgery, depending on the clinical features.[5]

SOLITARY FIBROUS TUMORS

Solitary fibrous tumors are exceedingly rare, accounting for 3.7% of all cases of soft-tissue sarcoma, with an incidence of 0.35 cases per 100,000 persons per year.[6] Such tumors can arise in the serous membranes, the dura of the meninges, and the deep soft tissues. Approximately 30% of cases arise in the pleura, 30% in the abdomen (including the peritoneum, retroperitoneum, and pelvis), 20% in the head and neck (including the meninges), and 20% in other sites (bone or deep soft tissues of the trunk, arms, or legs).[7] Patients with solitary fibrous tumors most commonly present with a palpable mass. The tumors harbor a characteristic genetic defect, inversion of the long arm of chromosome 12 (12q13), which results in fusion of *NAB2* (the gene that encodes next-generation fronthaul interface 1A–binding protein 2) and *STAT6* (the gene that encodes STAT6).[8-10] The growth pattern of solitary fibrous tumors varies widely, ranging from indolent to highly aggressive.

As an inpatient oncologist, when I evaluate a patient with metastatic cancer who presents with a complex medical problem such as hypoglycemia, I first try to determine whether the problem is caused by the cancer. In most cases, I find that the reason for the patient's hospitalization is related to the cancer, even if it might not seem that way at first. The possible cancer-related causes of this patient's recurrent hypoglycemia include an insulin-secreting tumor, a high metabolic rate with glucose consumption by the tumor, a side effect of therapy, or a paraneoplastic syndrome. The possible non–cancer-related causes are limited to surreptitious insulin use or sepsis. Sepsis might even be categorized as a cancer-related cause, given that patients with cancer are at an increased risk for severe infection due to neutropenia related to cytotoxic chemotherapy.

SURREPTITIOUS INSULIN USE OR SEPSIS

The key to determining the diagnosis in this case is to focus on the additional laboratory values obtained during the patient's second hospital admission, namely the insulin level (<0.2 μIU per milliliter), C-peptide level (<0.1 ng per milliliter), and β-hydroxybutyrate level

(<0.1 mmol per liter). The two possible non–cancer-related causes can be ruled out first. The patient had an undetectable insulin level, which rules out surreptitious insulin use. In addition, at the time of the current admission, he appeared well and had normal vital signs, a normal leukocyte count, and a normal lactate level (0.9 mmol per liter), findings that rule out sepsis as the cause of his hypoglycemia.

INSULIN-SECRETING TUMOR

Several cancer-related causes could explain this patient's hypoglycemia. An insulin-secreting tumor would result in a high insulin level, as well as a high C-peptide level.[11] As such, the C-peptide level can be used to differentiate an insulin-secreting tumor from exogenous insulin use. Both the insulin level and the C-peptide level were undetectable in this patient, so we can rule out insulinoma.

GLUCOSE CONSUMPTION BY THE TUMOR

Glucose consumption by a highly active tumor is an exceedingly rare phenomenon — essentially a spontaneous tumor lysis syndrome in a patient with a solid tumor. If this patient had a high metabolic rate with glucose consumption by the tumor, I would expect additional electrolyte disturbances to be present. The potassium and phosphorus levels would be high, and the calcium level would be low. The normal levels in this patient make this diagnosis unlikely.

SIDE EFFECT OF THERAPY

The use of glucocorticoids can lead to adrenal insufficiency. This patient had normal results on a cosyntropin stimulation test, a finding that rules out adrenal insufficiency. In addition, the use of immune checkpoint inhibitors can lead to the development of anti-insulin antibodies. He did not have exposure to immune checkpoint inhibitors, which rules out an immune-mediated adverse effect. Furthermore, if he had anti-insulin antibodies, I would expect the insulin level to be very high, and it was not.[12]

PARANEOPLASTIC SYNDROME

Oncologists consider the diagnosis of paraneoplastic syndrome in patients with cancer who have something going on that is difficult to explain. Some paraneoplastic syndromes are relatively common, such as the Lambert–Eaton syndrome in patients with small-cell lung cancer,[13,14] whereas other paraneoplastic syndromes are rare. I suspect that this patient had one of these unusual syndromes.

The characteristic genetic defect of solitary fibrous tumors is a chromosomal inversion that results in the *NAB2–STAT6* fusion gene. This fusion involves a constitutively activated version of *NAB2*, which drives tumorigenesis. In addition, the chimeric transcription factor produced by *NAB2–STAT6* causes dysregulation of several target genes, including *IGF2*, the gene that encodes insulin-like growth factor II (IGF-II).[8,15] Dysregulation of *IGF2* leads to excess production of IGF-II from the tumor, which can result in hypoglycemia.

In an article by Karl Walter Doege and Roy Pilling Potter that was published in the Annals of Surgery in 1930, the phenomenon of refractory hypoglycemia in a patient with "fibro-sarcoma" of the mediastinum was described for the first time.[16] This phenomenon has since been named the Doege–Potter syndrome and is known to be caused by tumor secretion of IGF-II. The Doege–Potter syndrome occurs in less than 5% of patients with solitary fibrous tumors, primarily with large peritoneal or pleural tumors.

If we refer once more to the patient's laboratory values, he had low levels of glucose, insulin, C-peptide, and β-hydroxybutyrate (a serum ketone). In the context of the Doege–Potter syndrome, these values would correlate with an increase in the free IGF-II level and an increased ratio of IGF-II to IGF-I. To establish the diagnosis of the Doege–Potter syndrome in this case, I would obtain the IGF-II level or the ratio of IGF-II to IGF-I.

CLINICAL IMPRESSION

Dr. Armen I. Yerevanian: This patient had symptomatic hypoglycemia that occurred both during fasting and in the postprandial period. It is important to note that during an episode of hypoglycemia (glucose level, 45 mg per deciliter [2.5 mmol per liter]), he had an insulin level of less than 0.2 μIU per milliliter and a C-peptide level of less than 0.1 ng per milliliter, findings that suggest non–insulin-mediated hypoglycemia.

Hypoinsulinemic hypoglycemia is a rare entity. Acute causes are critical illnesses associated with metabolic derangement, such as liver failure, kidney failure, adrenal crisis, or sepsis.[17] Congenital causes involve inborn errors of metabolism, such as glycogen storage diseases, fatty acid oxidation disorders, or mitochondrial diseases.[18] Acquired causes include insulin autoimmune syndrome and the overproduction of endogenous insulin paralogues, such as IGF-I and IGF-II.[19,20] In this patient, the solitary fibrous tumor and progressive hypoglycemia were suggestive of IGF-II overproduction.[21]

DR. TARA E. SOUMERAI'S DIAGNOSIS

Doege–Potter syndrome.

CLINICAL DIAGNOSIS

Hypoglycemia due to excess production of insulin-like growth factor II from a solid tumor (Doege–Potter syndrome).

DIAGNOSTIC TESTING

Dr. Albert L. Sy: Diagnostic testing was performed with the use of a competitive binding immunoassay.[22] The patient had an IGF-II level of 354 ng per milliliter (reference range, 333 to 967), an IGF-I level of 23 ng per milliliter (reference range, 50 to 317), and a glucose level of 60 mg per deciliter (3.3 mmol per liter; reference range, 70 to 110 mg per deciliter [3.9 to 6.1 mmol per liter]). In the context of paraneoplastic production of IGF-II, the ratio of IGF-II to IGF-I is the diagnostic marker of choice because the absolute IGF-II level is frequently normal.[23] A ratio of more than 3:1 is suggestive of tumor-mediated production of IGF-II, and a ratio of more than 10:1 is considered to be diagnostic.[24] This patient had a ratio of 15.4:1. The combination of a metastatic solitary fibrous tumor, hypoglycemia, and an IGF-II:IGF-I ratio of 15.4:1 is consistent with a diagnosis of the Doege–Potter syndrome.

Diagnostic testing for the Doege–Potter syndrome may include measurement of the pro-IGF-II level. Molecular testing for confirmation of a *NAB2–STAT6* fusion gene may be performed but is not required; a positive immunohistochemical stain for STAT6 confirms the diagnosis.

PATHOLOGICAL DIAGNOSIS

Doege–Potter syndrome.

DISCUSSION OF MANAGEMENT

Dr. Yerevanian: Solitary fibrous tumors and the associated *NAB2–STAT6* fusion gene drive the expression of the IGF-II precursor protein known as pro-IGF-II (Fig. 3).[8] Post-translational products of pro-IGF-II (known as "big IGF-II") can occur at high free levels because of reduced binding to IGF-binding proteins.[25-27] Circulating IGF-II can bind to the insulin receptor directly or to heterodimers of the insulin receptor and the IGF-II receptor, inducing peripheral glucose internalization and reduced hepatic glucose output in the absence of insulin.[28]

The definitive treatment for hypoglycemia due to IGF-II overproduction is the elimination of excess production of IGF-II through treatment of the primary cancer. However, if the cancer cannot be resected, the goal of treatment is symptom control. Mild hypoglycemia can be managed with frequent carbohydrate intake and continuous glucose monitoring.[29] This patient was advised to start having frequent meals, including carbohydrate intake at night. Despite the use of this strategy, the patient had persistent hypoglycemic episodes.

The use of high-dose glucocorticoids (the equivalent of >20 mg of prednisone per day) can reduce the production of IGF-II in the tumor, alter the levels of IGF-binding proteins, and promote resistance to insulin-receptor signaling.[30,31] We treated this patient's refractory hypoglycemia with prednisone administered in the evening. As his tumor burden worsened, the prednisone dose was increased from 30 mg per day to 60 mg per day in split doses. Despite the use of these high doses of glucocorticoids, the patient continued to have breakthrough hypoglycemia.

Recombinant human growth hormone has been shown to reduce hypoglycemia through suppression of hepatic insulin signaling, increased production of IGF-binding protein 3, and reduction of the bioavailability of circulating IGF-II.[32,33] Somatostatin analogues have not been shown to have a benefit with respect to reducing hypoglycemic episodes.[34] A high tumor burden and persistent hypoglycemia are indications for palliative treatment with glucagon infusions, dextrose infusions, or parenteral nutrition.[35] Investigational agents such as IGF-II antagonists have been evaluated but are not yet available for clinical use.[36,37]

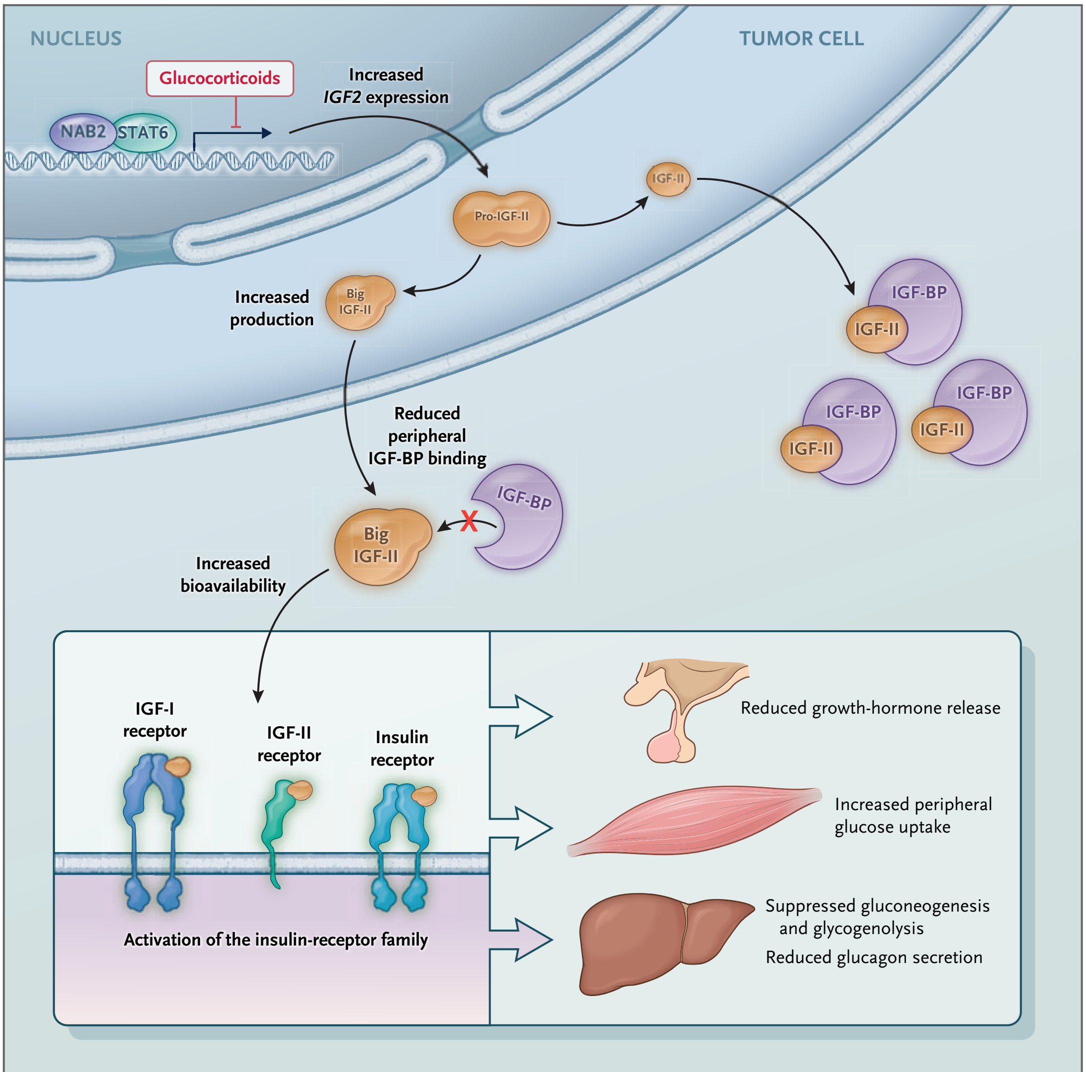

Figure 3. Pathophysiology of the Doege–Potter Syndrome in Patients with Solitary Fibrous Tumors.

Solitary fibrous tumors are frequently associated with the presence of the *NAB2–STAT6* fusion gene. The chimeric protein (NAB2–STAT6) produced from this chromosomal rearrangement is strongly translocated to the nucleus and drives expression of *IGF2*, the gene that encodes insulin-like growth factor II (IGF-II). The precursor polypeptide of IGF-II is differentially processed in these tumors, with a substantial increase in excretion of the 1–87 amino acid product of pro-IGF-II, known as “big IGF-II.” As compared with IGF-II, big IGF-II does not bind as tightly to circulating IGF-binding proteins (IGF-BPs), which increases its bioavailability and signaling capacity. The increase in IGF-II products promotes systemic activation of the insulin-receptor family, including the IGF-I receptor, the IGF-II receptor, and the insulin receptor. Signaling in the central nervous system reduces the release of growth hormones through a negative-feedback mechanism. Enteric signaling suppresses gluconeogenesis and glycogenolysis and reduces glucagon secretion. Systemic signaling promotes glucose uptake and induces the hypoglycemia that is characteristic of the Doege–Potter syndrome. Glucocorticoids directly reduce the production of tumor-derived IGF-II isoforms and also reduce the systemic effects of insulin signaling.

Dr. Cote: The initial treatment for a solitary fibrous tumor is dictated by clinical features, such as staging (size, location, and metastasis), pathological risk stratification, and patient-specific factors. For a localized solitary fibrous tumor, treatment involves en bloc surgical resection with wide negative margins, when feasible. For a higher-risk tumor, radiation therapy may be added before or after surgery.[38-43] For an unresectable solitary fibrous tumor, definitive radiation therapy may be considered.[44]

Metastatic solitary fibrous tumors are typically not curable, and therefore, systemic anticancer therapy is often used to increase survival and palliate symptoms. Because this cancer is rare, there is a paucity of data from randomized trials to guide treatment. Decisions regarding systemic therapy are based on data from small single-group studies, case series, or anecdotal reports. Therapies with reported activity include alkylating agents (e.g., dacarbazine[45] and temozolomide), anthracyclines (e.g., doxorubicin), and antiangiogenic agents (e.g., sunitinib,[46] pazopanib,[47] and axitinib[48]). Some may be used in combination, such as dacarbazine with doxorubicin or temozolomide with bevacizumab.[49,50] Given the relatively low efficacy of these therapies, we strongly encourage patients to consider participation in clinical trials, when available. In addition, for patients who have a differential response to treatment (with one metastasis progressing and other metastases remaining stable), oligometastatic disease, or paraneoplastic syndromes, we consider techniques such as radiation therapy, ablation therapy, or surgical resection or debulking.

This patient had an aggressive solitary fibrous tumor with a difficult clinical course. He was treated with multiple anticancer agents, and despite these attempts, his cancer progressed. Two months after the hospitalization, in the context of uncontrolled cancer symptoms, the patient was transitioned to comfort measures and died peacefully with his family at the bedside.

FINAL DIAGNOSIS

Doege–Potter syndrome.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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